

Group Office Hours - February 22, 2021

ECON 301

Spring 2021



Task 5

$$(d) \quad \text{Var}(Y_i | X_1) = \text{Var}(Y_j | X_j) \quad \text{for all } j = 1, \dots, n$$

$$\text{Cov}(Y_i, Y_j | X_1, \dots, X_n) = \underbrace{E[Y_i Y_j | X_1, \dots, X_n]}_{f(y_i, y_j | X_1, \dots, X_n)} - \underbrace{E[Y_i | X_1, \dots, X_n]}_{f(y_i | X_1, \dots, X_n)} \underbrace{E[Y_j | X_1, \dots, X_n]}_{f(y_j | X_1, \dots, X_n)}$$

$$f(y_i, y_j | X_1, \dots, X_n)$$

$\Downarrow$ 5c

$$= f(y_i | X_i) f(y_j | X_j)$$

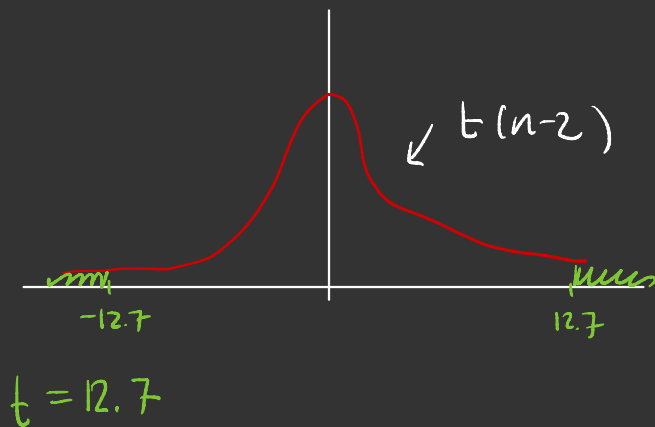
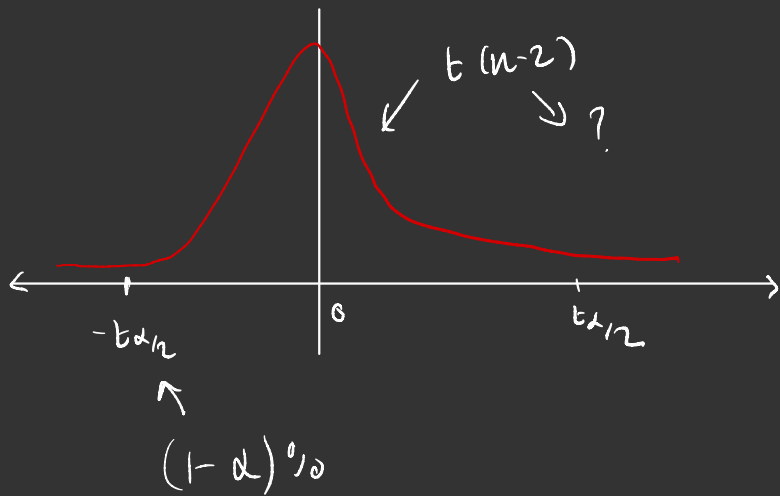
$$f(y_i | X_1, \dots, X_n)$$

$\Downarrow$ 5b

$$= f(y_i | X_i)$$

$$E[Y_i Y_j | X_1, \dots, X_n] = \int \int y_i y_j f(y_i | X_i) f(y_j | X_j) dy_i dy_j = \begin{cases} \text{if } i=j & E[Y_i^2 | X_i] \\ \text{if } i \neq j & E[Y_i | X_i] E[Y_j | X_j] \end{cases}$$

fill in
the gaps.



$$\text{Var}(\bar{X} | X's) = \bar{X}^2 \text{Var}(1 | X's) = 0$$

↑
Thm 2.2.

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i = f(X_1, \dots, X_n) \\ = f(X's)$$

$$\text{Var}(\bar{X} \hat{\beta}_1 | X's) = \bar{X}^2 \text{Var}(\hat{\beta}_1 | X's) = \bar{X}^2 \frac{\sigma^2}{S_{xx}}$$

$$\left[\begin{array}{l} \text{A4 homoskedasticity} \\ \text{Var}(Y|X) = \sigma^2 \end{array} \right]$$

A1 - A4

$$\sum_{i=1}^3 (X_i - \bar{X}) \left(\frac{3X_i + 2}{6X_i + 3} \right)$$

$$\sum_{i=1}^n (X_i - \bar{X}) = 0$$

$$\sum_{i=1}^n a_i (X_i - \bar{X}) \neq 0$$